

Georg von Westphalen

MEDKAYFAB: ENDOCRINOLOGY EDITION

Uncle FrizzleBob's Krazy Kayfabizzaro Komik Kard Kult Klub
for Narrative Medical Learning Games!



STAT BLOCK: ENDOCRINE SYSTEM - HORMONE
REGULATION, METABOLISM, GROWTH,
RESPONSE TO STRESS, HOMEOSTASIS.

KEY GLANDS: PITUITARY, THYROID,
ADRENAL, PANCREAS, GONADS.

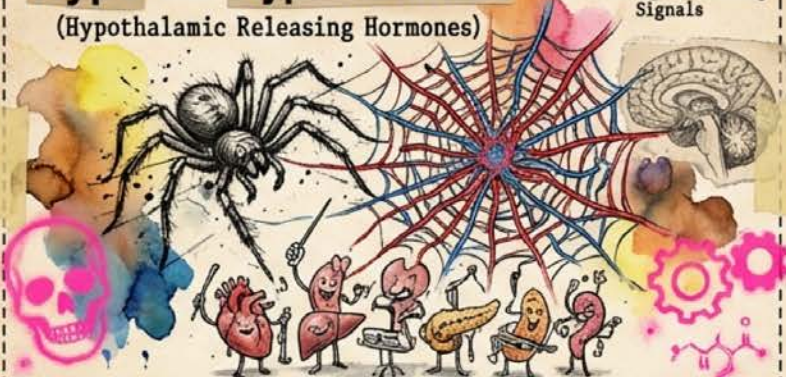
UNCLE FRIZZLEBOB SAYS: "IT'S ALL
ABOUT THE CHEMICAL CHAOS, BABY!"

KayfaBIZARRO!
Burn your textbooks!
Stay fluffy! →

Hypo the Hypothalamus

(Hypothalamic Releasing Hormones)

STAT: 5
Releasing/Inhibiting
Signals



POWER: Portal Command - Send short-range chemical orders via blood, bypassing nerves.
LORE: Dopamine tonically inhibits prolactin. Metoclopramide blocks dopamine → galactorrhea.

GH the Giant

(Growth Hormone)

STAT: IGF-1
(Best Screening Marker)



POWER: Paradoxical Rise - Falls to suppress during oral glucose tolerance test (OGTT).
LORE: Acromegaly causes insulin resistance. Diagnose with high IGF-1 and failure to suppress GH on OGTT.

Piti the Anterior Pituitary

(Anterior Pituitary Hormones)

STAT: 6
Major Hormones
(FLAT PIG)



POWER: Middle Management - Amplify hypothalamic orders, but yield to peripheral feedback.
LORE: A large non-functioning adenoma causes mass effect → bitemporal hemianopia + secondary hypopituitarism.

Proolly the Prolactin

(Hyperprolactinaemia)

STAT:
Prolactin >100 µg/L
(Stalk Compression)



POWER: Tonic Suppression - Unique hormone controlled by constant INHIBITION.
LORE: Stalk compression blocks dopamine → mild prolactin rise. Treat true prolactinomas with dopamine agonists (cabergoline).

CHAGS! PUNK MEDICINE DIY OR DIE HORRORFOL ANARCHY CUT & PASTE MIND OVER MATTER

GUTTER SCIENCE INK BOMBS READ ZINES!

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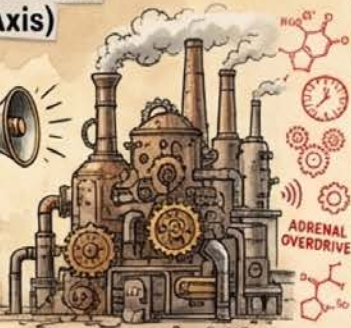
ACTH the Commander (HPA Axis)

STAT: Peaks at 8 AM

CORTISOL RUSH!



WAKE UP!



POWER: Circadian Assault - Drive cortisol production, losing rhythm in Cushing's.

LORE: High ACTH + High Cortisol = ACTH-dependent Cushing's.

Low ACTH + High Cortisol = Adrenal adenoma.

ADH the Water Keeper (Vasopressin)

STAT: Osmolality Threshold 280 mOsm/kg

AQUAPORIN!



PEE LESS!



POWER: V2 Insertion - Retain free water in the collecting duct.

LORE: Central DI lacks ADH (treat with desmopressin).

Nephrogenic DI resists ADH (treat with low-salt diet/thiazides).

Thyro the Thyroid Axis (HPT Axis)

STAT: T3 is 3-5x more potent than T4

METABOLISM BOOST!



POWER: Deiodinate Conversion - Convert T4 to the highly active T3 in peripheral tissues.

LORE: Pregnancy increases TBG → total T4 rises, but free T4 is normal. Do not adjust dose based on total T4!

Iggy IGF-1 (GH Mediator)

STAT: Long Half-Life (Stable Marker)



POWER: Prexy Integration - Provide a stable readout of erratic, pituitary GH secretion.

LORE: IGF-1 falls in malnutrition even if GH is high (GH resistance). Somatostatin analogues suppress GH, reducing IGF-1.



Cortisol the Stress Master

(Glucocorticoids)

STAT:
<50 nmol/L
(Normal 1mg DST
suppression)



POWER: Emergency Fuel – Mobilize glucose at the cost of muscle, bone, and immune function.
LORE: Exogenous steroids suppress the HPA axis. Abrupt withdrawal during infection causes fatal adrenal crisis.

Thyra Synthesis

(Thyroid Assembly)

STAT: 37% Iodine
(Amiodarone weight)



POWER: Wolff-Chaikoff Effect – Massive iodine loads shut down synthesis.
LORE: PTU blocks synthesis AND peripheral T4→T3 conversion. Preferred in 1st trimester pregnancy and thyroid storm.

Feedback Freda

(Negative Feedback)

STAT: 2
Diagnostic
Directions



POWER: Axis Algorithm – High peripheral hormone + low tropic hormone = autonomous peripheral gland.
LORE: Use stimulation tests (ITT, Synacthen) for suspected deficiency.
Use suppression tests (OST, OGTT) for suspected excess.

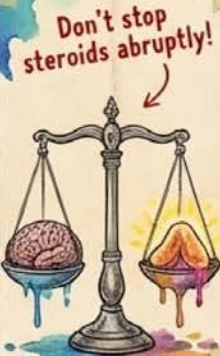
Hashi the Destroyer

(Hashimoto's)

STAT: 95%
Anti-TPO
positive



POWER: Lymphocytic Infiltration – Slowly run the body at half-speed via Th1 destruction.
LORE: Severe hypothyroidism causes myxoedema (non-pitting oedema) and delayed relaxation of ankle reflexes.



READ ZINES!
PUNK MEDICINE
GIVE STITCHES
READ ZINES!
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CHAOS! PUNK MEDICINE DIY or DIE HOBIOTICAL ANARCHY CUT & PASTE MIND OVER MATTER

GOTTER SCIENCE INK BOMBS READ ZINES!

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READ ZINES!

KoyfaBINGO! The orids wilness!

GRAVES the Impersonator (Autoimmune Hyperthyroidism)

STAT: 70-80% of
Hyperthyroidism



POWER: Constitutive Activation – TSI permanently overrides feedback control.
LORE: TSH-R on orbital fibroblasts causes exophthalmos, which persists even after thyroid treatment.

THYROID STORM TOM (Life-Threatening Thyrotoxicosis)

STAT: 10-30%
Mortality



POWER: Supercritical Melt – Acute physiological stress triggers lethal high-output failure.
LORE: You **MUST** give PTU before Lugol's iodine, or the iodine load will flood out as new hormone.

NODULE NORMAN (Thyroid Nodules)

STAT: 50-70%
Prevalence
on US



POWER: The TSH Fork – Low TSH = hot/autonomous (rarely malignant).
Normal/High TSH = Needs US/FNA.
LORE: Papillary cancer (80%) spreads via lymph nodes. Follicular cancer (15%) spreads via blood.

HYPO the Hypothyroid Drug (Levothyroxine)

STAT: 6 Weeks
to Steady
State



POWER: Prodrug Conversion – Relies on peripheral deiodinase; takes weeks to reflect dose changes.
LORE: Absorption is impaired by food, iron, calcium. Dose must increase 25-50% immediately upon pregnancy confirmation!

HORMONAL
REVOLUTION!



What the
FLUFF?!
Orange-peel
Vial

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THYROID
CHAOS!



THYROID
REVOLT!

CHAOS
FACTOR

KayfaBOGGLE!

Parathyroid Pete (PTH)

STAT: 4 Glands, Rice-Sized



POWER: Calcium Reclamation - High Ca, Low PO_4 , High urine Ca.

LORE: Primary hyperparathyroidism is the most common outpatient cause of hyperca of hypercalcaemia.

MURPHY

Vitamin D the Sunshine Hormone (Metabolism)

STAT: $25(OH)D_3 > 50 \text{ nmol/L}$



POWER: Dual Hydroxylation - Must be activated sequentially to absorb gut calcium.

LORE: Sarcoidotic macrophages have 1α -hydroxylase activity, causing unregulated active calcitriol and hypercalcaemia.

Addie the Adrenal Architecture (Adrenal Zones)

STAT: GFR = Salt, Sugar, Sex



POWER: Triple Hormone Threat - Addison's destroys all three cortical layers.

LORE: Zona Glomerulosa makes Aldosterone (regulated by RAAS/ K^+ , NOT ACTH). Fasciculata makes Cortisol. Reticularis makes Androgens.

MEDULLA (Adrenaline)

Aldo the Aldosterone (Mineralocorticoid)

STAT: ARR (Aldosterone:Renin Ratio)



POWER: Volume Controller - $ENaC$ insertion raises pressure at the cost of potassium.

LORE: Primary hyperaldosteronism (Conn's) causes hypertension + hypokalaemia with SUPPRESSED renin.

Bones, stones, groans!

GW

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Cushing the Cushioned (Cortisol Excess)

STAT: 70% Pituitary (Cushing's Disease)



POWER: Ectopic Overdrive – SCLC ectopic ACTH causes profound hypokalaemia, rapid onset, less classic appearance.

LORE: Screen with 24h urinary free cortisol, late-night salivary cortisol, or low-dose dexamethasone suppression test.

Addison the Insufficient (Primary AI)

STAT: 80% Autoimmune (Developed World)



POWER: MC1R Stimulation – High ACTH cross-reacts to cause hyperpigmentation (palmar creases, buccal mucosa).

LORE: Secondary adrenal insufficiency has LOW ACTH, therefore NO hyperpigmentation.

Pheo the Hidden (Phaeochromocytoma)

STAT: 25-30% Hereditary (MEN2, VHL, NF1)



POWER: Paroxysmal Spells – The classic triad: headache, sweating, palpitations.

LORE: Never give beta-blockers without alpha-blockade first; unopposed alpha causes lethal paradoxical hypertension!

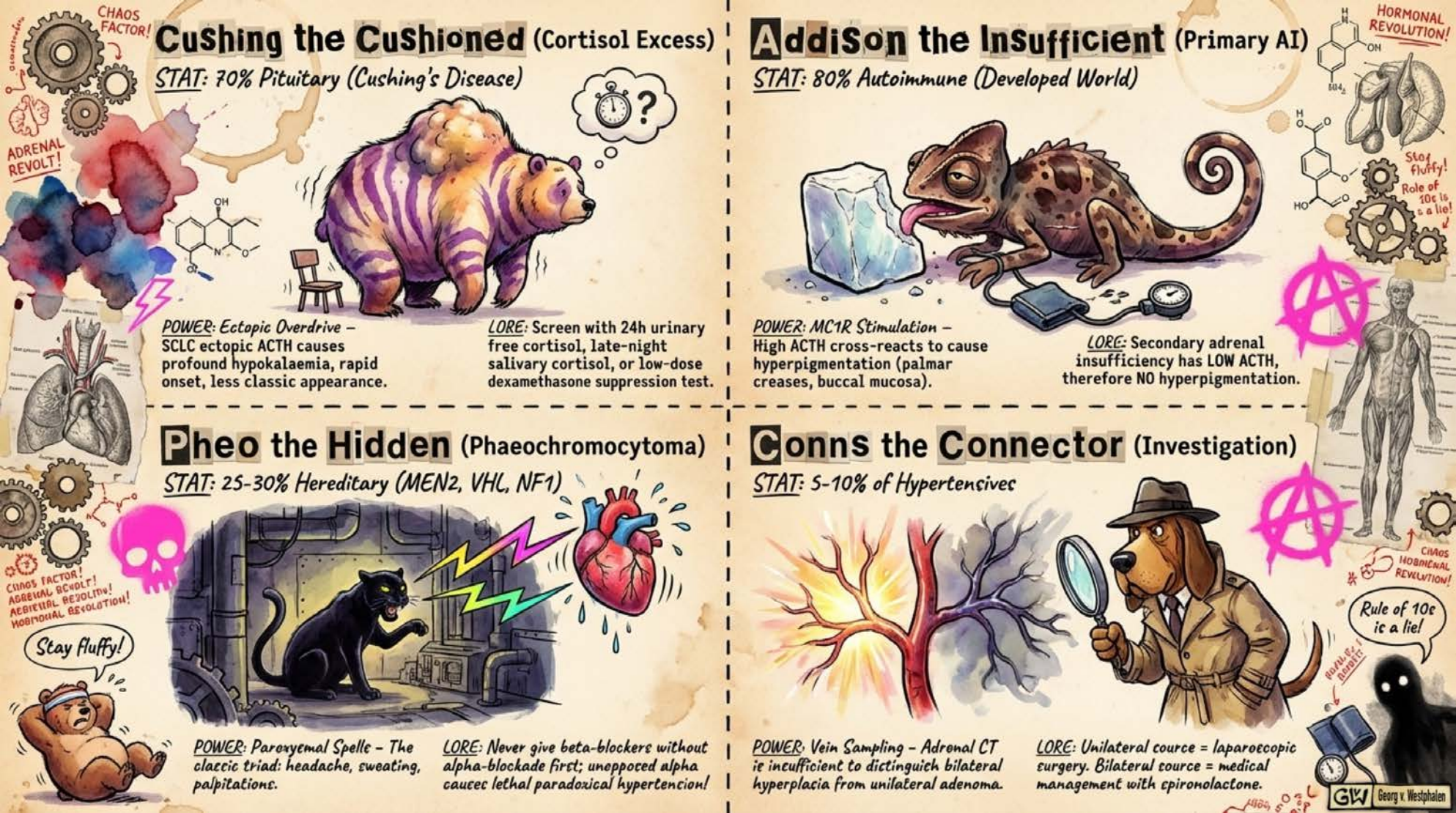
Conns the Connector (Investigation)

STAT: 5-10% of Hypertensives



POWER: Vein Sampling – Adrenal CT is insufficient to distinguish bilateral hyperplasia from unilateral adenoma.

LORE: Unilateral course = laparoscopic surgery. Bilateral course = medical management with eplerenone.



CAH Clara (21-Hydroxylase Deficiency)

STAT: 90% of CAH



POWER: Enzyme Blockade – Lack of cortisol removes negative feedback → high ACTH drives massive androgen overproduction.

LORE: Classic salt-wasting causes neonatal crisis. Non-classic mimics PCOS in adults. Diagnose with elevated 17-OHP.

Waterhouse the Crisis (Adrenal Crisis)

STAT: 100mg IV Hydrocortisone



POWER: Sick Day Rules – Double or triple oral steroid dose during fever; IM/IV

LORE: Do NOT wait for lab results. Give 100mg IV hydrocortisone + large volume 0.9% NaCl immediately for unexplained hypotensive shock.

MEN the Syndrome Families (MEN1, 2A, 2B)

STAT: 3 Ps (MEN1)



POWER: Genetic Destiny – RET proto-oncogene mutations dictate medullary thyroid cancer prophylactic resection.

LORE: MEN2B = Medullary Thyroid Ca, Phaeo, Mucosal Neuromas, NO parathyroid.

Incidentaloma Ida (Adrenal Incidentaloma)

STAT: 2–3% of Adult CT Scans



POWER: Dual Threat Matrix – Is it hormonally active? AND Is it malignant (>4cm, >10 Hounsfield Units)?

LORE: You MUST exclude pheochromocytoma via plasma metanephrines before attempting any adrenal biopsy, or risk a fatal hypertensive crisis.

TIME BOMB IN A GLAND!

Insu the Insulin

(Secretion & Action)

STAT: 51 Amino Acids



POWER: Biphasic Release - Cleaves C-peptide from proinsulin to unlock the body's only glucose-lowering mechanism.

LORE: Endogenous insulin = high C-peptide. SurpeEtious exogenous insulin injection = undetectable C-peptide.

Gluca the Glucagon

(Counter-Regulatory)

STAT: Hepatic Action Only



POWER: Glycogenolysis - Mobilizes hepatic stores within minutes to fight severe hypoglycaemia.

LORE: Glucagon rescue injections fail in alcoholic or prolonged starvation because hepatic glycogen stores are already depleted.

T1D the Autoimmune (Type 1 Diabetes)

STAT: Anti-GAD65 (Most Sensitive)



POWER: Absolute Deficiency - Without insulin, the body metabolizes itself into fatal acidosis.

LORE: DKA occurs readily in T1D because zero residual insulin means unopposed lipolytic. T2D rarely develops DKA due to residual insulin.

T2D the Resistance (Type 2 Diabetes)

STAT: 5-10% Weight Loss = Magic



POWER: The Slow Siege - Visceral fat drives resistance → hypersecretion → beta cell exhaustion.

LORE: Metformin is first line. Add SGLT2i or GLP-1 RA if established ASCVD, Heart Failure, or CKD independent of glucose



Metty the Metformin

(Mechanism)

STAT: eGFR <30
(Contraindicated)

KayfaBONGO!
Protect the heart!



POWER: Weight Neutrality - Lowers fasting glucose without rickng hypoglycaemia.
LORE: Hold metformin 48h for iodinated contract; contract-induced AKI causes drug accumulation, leading to rare but fatal lactic acidosis.

GLP-1 Gloria

(Receptor Agonists)

STAT: Up to 15%
Body Weight Loss



POWER: Glucose-Dependent Satiety - Only releases insulin when glucose is high = low hypo risk.
LORE: Unmatched for weight loss and cardiovascular MACE reduction, but causes dose-dependent GI nausea and delays gastric emptying.

SGLT2 Stan

(SGLT2 Inhibitors)

STAT: -0.5 to 1% HbA1c



POWER: Osmotic Diuretic - Reduces HF hospitalizations and CKD progression independent of glucose.
LORE: Can cause euglycaemic DKA (ketones high, glucose normal) and genital mycotic infections. Stop before surgery!

DPP-4 Dave

(Gliptins)

STAT: Prolongs
 $t_{1/2} > 2$ mins



POWER: Enzyme Blockade - Modest oral incretin effect, weight neutral, safe.
LORE: Produces less effect than GLP-1 RAs because it only prevents degradation of physiological levels, rather than providing pharmacological super-doses.

SULFO the Sulfonylurea

(Gliclazide/Glibenclamide)

STAT: SUR1 Receptor Target



POWER: Glucose-Independent Release – High risk of severe hypoglycaemia + weight gain.

LORE: Avoid renally cleared glibenclamide in the elderly; metabolites accumulate causing prolonged, dangerous hypoglycaemia.

DKA the Crisis

(Diabetic Ketoacidosis)

STAT: pH <7.3, Ketones >3



HUMBUG!
Don't drop the osmolality too fast!

POWER: FRIII Protocol – 0.1 units/kg/hr fixed rate insulin infusion.

LORE: Total body potassium is depleted despite presenting hyperkalaemia. Insulin drives K⁺ back into cells; you MUST add KCl to IV fluids!

HHS the Hyperosmolar

(Hyperglycaemic State)

STAT: Osmolality >320, Mortality 15-20%



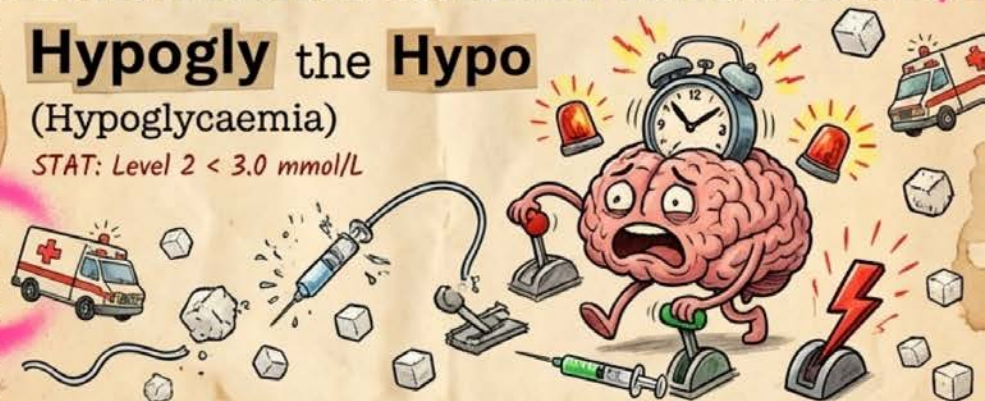
POWER: Profound Dehydration – Residual insulin prevents ketosis, allowing glucose to rise unchecked.

LORE: Replace fluids slowly (0.9% NaCl) to avoid cerebral oedema. Do NOT start insulin immediately unless glucose falls to fall with fluids alone. LMWH is essential!

Hypogly the Hypo

(Hypoglycaemia)

STAT: Level 2 < 3.0 mmol/L

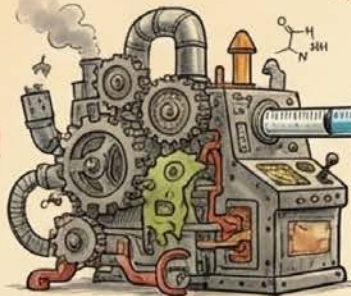


POWER: Autonomic Warning – Sweating and tremor precede neuroglycopenic coma.

LORE: Recurrent hypos cause Hypoglycaemia Unawareness—autonomic failure where the threshold drops and warning symptoms are lost

Insulinoma Ivan (Beta Cell Tumor)

STAT: 72-Hour
Supervised Fast

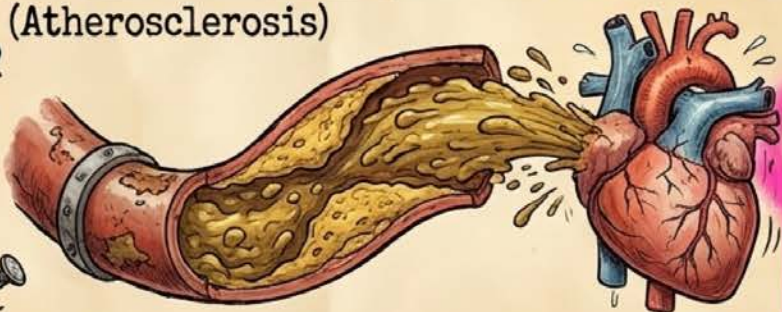


What the
FLUFF?!
Glove and
stocking!

Whipple's Triad – Hypo symptoms + documented low glucose + relief with carbs.
LORE: Endogenous hyperinsulinism shows High Insulin + High C-Peptide + Low beta-hydroxybutyrate (insulin suppresses ketogenesis).

Macrovascular Max (Atherosclerosis)

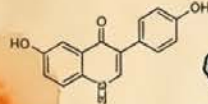
STAT: #1 Cause
of Death in T2DM



POWER: Endothelial Dysfunction – Diabetes triples ASCVD risk.
LORE: Glucose control prevents microvascular disease, but macrovascular protection requires high-intensity Statins, ACEI, and SGLT2/GLP-1 RAs.

Complications Combo (Microvascular)

STAT: Annual
Retinal/Foot
Screen



POWER: AGEs & Polyol Pathway – Chronic glycation destroys small vessels and pericytes.
LORE: Loss of pain sensation (neuropathy) + repeated micro-trauma = painless foot ulcers and the radiologically destroyed Charcot joint.

The Diabetes Attending (Master Review)

STAT: 3 Simultaneous Cases



POWER: Synthesis Challenge – DKA electrolytes vs. eGFR 28 drug limits vs. Fasting Hypo localization.
LORE: "Map every decision to its pharmacological, physiological, or anatomical basis. No gestalt. Every answer needs a mechanism."

Osteo the Osteoporosis

(Bone Loss)

STAT: T-score ≤ -2.5



POWER: Silent Collapse - Fracture threshold crossed without symptoms.

LORE: Denosumab stops osteoclasts, but stopping it causes massive rebound fractures. Never stop without transitioning to a bisphosphonate!

KayfaBIZARRO!
Rebound fractures!

RANKL the Bone Regulator

(Signalling)

STAT: RANKL : OPG Ratio



POWER: Immune-Bone Coupling - Why inflammation melts bone.

LORE: Rheumatoid arthritis causes periarticular erosion via RANKL; anti-TNF biologics and Denosumab block this exact destruction pathway.

Hyper Calcium Harry

(Hypercalcaemia)

STAT: Ca > 3.5 = Emergency



POWER: End-Organ Calcification - Bones, stones, groans, psychic moans.

LORE: Primary hyperparathyroidism = High PTH, High Ca. Malignancy = Low PTH, High Ca, High PTHrP. Treat severe with 3-4L 0.9% NaCl + IV Zoledronate.

KayfaBIZARRO!
Rebound fractures!

Hypo Calcium Hettie

(Hypocalcaemia)

STAT: Tap Facial Nerve (Chvostek's)



POWER: Neuronal Trigger - Low calcium drops the threshold for action potentials $\rightarrow$ tetany.

LORE: Hypomagnesaemia impairs PTH release AND action. Refractory hypocalcaemia will not fix with calcium alone; you must check and replace GW Georg v. Westphalen

PAGET THE BONE REMODELLER

(Paget's Disease)

STAT1: Isolated highly elevated ALP



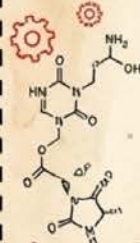
POWER: Focal Chaos - Parasyaovirus-driven osteoclast hyperactivity followed by disorganized repair.

LORE: Massively enlarged skull bone compresses the 8th cranial nerve, causing hearing loss. Treat with IV Zoledronate.

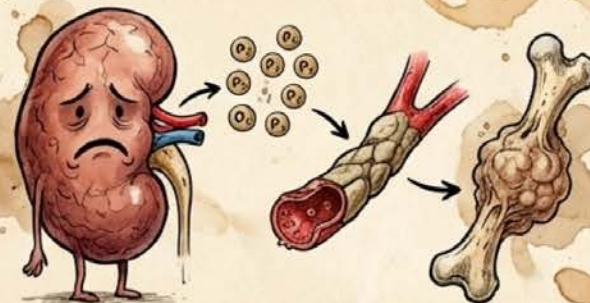
RENAL BONE RENÉE

(CKD-MBD)

STAT1: High POF-23



Stay floppy! Water, water everywhere!



POWER: The Fatal Cascade - High PO_4 → Low Calcitriol → Low Ca → Secondary Hyperparathyroidism.

LORE: Prolonged parathyroid stimulation becomes autonomous ('Tertiary'), maintaining high PTH even after a renal transplant!

DIABETES INSIPIDUS DANA

(ADH Defects)

STAT1: Urine Osmolality <300



POWER: Water Deprivation - Fails to concentrate urine when dehydrated.

LORE: Central DI responds to desmopressin. Nephrogenic (Lithium) does not. Paradoxically, thiazide diuretics reduce urine output in Nephrogenic DI!

SIADH SALLY

(Euvolaemic Hyponatraemia)

STAT1: Urine Na > 20, Plasma Na < 135



POWER: Inappropriate Retention - Small Cell Lung Cancer (ectopic ADH) secretly diluting the blood.

LORE: Correcting sodium too fast (>8-10 mmol/L/24h) causes Osmotic Demyelination Syndrome (locked-in syndrome). Restrict fluids first!

KayfaBONGO!

You survived the Kult!



SECRETIN PARADOX!



Thyroid Cancer Tracy

STAT: 80% Papillary (Best Prognosis)

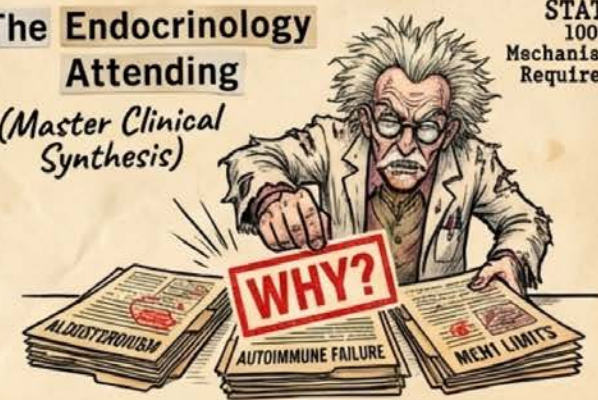


POWER: Histological Destiny - Cytology dictates surgical and radioactive iodine strategy.
LORE: Follicular carcinoma is impossible to diagnose on FNA; you need whole histology to see capsular/vascular invasion!

The Endocrinology Attending

STAT: 100% Mechanism Required

(Master Clinical Synthesis)



POWER: Cross-Axis Integration - Primary hyperaldosteronism lateralization vs. Autoimmune polyglandular failure vs. MEN1 surgical limits.
LORE: "Name the diagnosis, explain the mechanism, complete the investigation, and prescribe the treatment. Gestalt is dead."

MEN1 Marco

STAT: Fasting Gastrin > 10x Normal



POWER: Secretin Paradox - Secretin normally inhibits gastrin, but causes a paradoxical MASSIVE RISE in gastrinoma.
LORE: MEN1 gastrinomas are multicentric. Stop PPIs before testing fasting gastrin, as PPIs artificially elevate it in everyone!

DIY Epistemic Sabotage

STAT: ∞

(Create Your Own)



POWER: Narrative Mnemonic - You remember stories, not lists.
LORE: Draw your own damn disease! If you can map the mechanism to a visual metaphor, you will never forget it on the wards. Stay Fluffy!

